

BOULDER, NV, USA

WMO: 720741

Lat: 35.947N

Lon: 114.861W

Elev: 2202

StdP: 13.56

Time Zone: -8.00 (NAP)

Period: 10-19

WBAN: 00269

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	31.7	34.0	-3.7	4.9	56.1	0.3	6.0	57.0	26.4	51.3	24.4	49.6	3.7	340	0.482

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.5	108.5	64.8	106.2	64.7	104.0	64.6	71.6	92.5	70.5	91.8	69.6	91.1	10.8	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.1	104.5	78.4	63.9	96.5	80.3	62.6	92.1	82.7	36.7	92.6	35.8	91.0	34.9	90.2	74.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.1	23.3	20.0	DB	27.0	112.2	2.5	1.6	25.2	113.3	23.7	114.3	22.3	115.2	20.5	116.3	(4)
(5)			WB	21.4	73.6	2.4	0.9	19.6	74.2	18.2	74.8	16.8	75.3	15.0	75.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	68.9	49.2	52.8	60.7	67.6	74.0	86.9	90.6	89.0	81.9	68.9	56.6	47.6	(6)	
	DBStd	16.29	5.85	7.12	6.58	7.03	7.54	5.75	4.49	4.43	6.03	6.67	7.01	5.60	(7)	
	HDD50	269	82	51	4	1	0	0	0	0	0	1	19	111	(8)	
	HDD65	1902	490	343	163	53	16	0	0	0	0	38	261	538	(9)	
	CDD50	7167	57	129	337	529	744	1106	1258	1208	957	587	217	38	(10)	
	CDD65	3324	0	1	31	131	294	656	793	743	507	159	9	0	(11)	
	CDH74	49473	0	27	412	1537	3824	10696	13063	11596	6655	1548	115	0	(12)	
	CDH80	30668	0	0	98	582	1934	7136	8929	7610	3841	533	5	0	(13)	
(14) Wind	WSAvg	7.0	5.6	6.8	7.9	8.9	8.6	8.5	7.7	6.8	6.2	6.3	5.8	5.5	(14)	
(15) Precipitation	PrecAvg	6.0	0.6	0.7	0.7	0.3	0.2	0.1	0.4	0.8	0.6	0.3	0.6	0.4	(15)	
	PrecMax	13.3	2.9	3.1	5.0	3.0	1.4	0.7	2.1	5.1	3.7	2.2	2.6	2.4	(16)	
	PrecMin	1.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)	
	PrecStd	2.7	0.7	0.8	1.0	0.6	0.3	0.2	0.6	1.1	0.9	0.5	0.7	0.6	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	69.8	77.3	87.8	93.5	100.4	112.6	111.2	108.5	104.1	93.1	81.0	68.2	(19)	
		MCWB	49.9	52.4	55.7	57.2	59.6	64.5	66.4	65.2	65.3	61.3	54.7	50.2	(20)	
	2%	DB	65.8	73.1	82.2	90.5	97.5	108.5	108.7	106.4	100.5	89.9	77.2	64.1	(21)	
		MCWB	47.2	50.6	53.7	56.3	58.4	63.7	65.5	65.1	64.1	59.3	52.7	48.1	(22)	
	5%	DB	63.2	70.4	78.9	86.3	93.4	105.9	106.3	104.2	99.2	85.8	73.5	61.3	(23)	
		MCWB	45.8	49.7	51.6	55.1	56.9	62.8	65.9	65.2	63.6	57.8	51.2	46.0	(24)	
	10%	DB	60.6	66.1	74.8	82.2	90.5	102.3	103.9	102.0	96.8	82.1	70.2	58.5	(25)	
		MCWB	44.6	47.8	50.4	53.3	56.3	61.2	65.9	65.2	63.0	56.0	50.2	45.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.1	56.4	56.8	59.1	61.5	68.9	72.9	73.1	71.5	66.3	56.9	54.7	(27)	
		MCDB	60.7	70.0	80.5	88.7	91.7	101.9	93.9	92.2	85.6	77.8	76.1	60.4	(28)	
	2%	WB	51.0	53.1	55.0	57.4	59.8	66.4	71.7	71.6	69.4	63.3	54.7	52.1	(29)	
		MCDB	58.1	66.1	77.3	84.2	91.0	99.1	93.1	92.6	86.8	82.3	73.9	59.3	(30)	
	5%	WB	49.3	51.1	53.4	56.0	58.6	64.7	70.8	70.5	68.2	61.1	52.7	50.0	(31)	
		MCDB	57.8	64.6	74.1	81.1	88.9	99.2	92.7	91.6	87.0	77.8	71.2	58.2	(32)	
	10%	WB	47.2	49.3	51.8	54.5	57.2	63.0	69.9	69.5	67.0	59.1	50.8	47.4	(33)	
		MCDB	57.0	65.3	72.3	78.8	86.0	97.9	92.3	90.5	87.0	76.0	68.2	55.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	18.8	20.8	22.8	23.7	24.3	27.9	24.5	24.5	24.5	22.8	21.8	17.9	(35)	
		MCDBR	22.7	26.3	28.4	29.6	28.8	29.7	27.5	27.7	27.2	26.1	25.7	22.2	(36)	
	5% WB	MCWBR	11.8	11.8	11.2	10.7	9.4	8.3	7.4	8.1	9.3	9.6	11.6	11.7	(37)	
		MCWBR	17.5	22.0	25.1	25.9	26.5	26.2	20.2	20.1	21.9	19.9	23.8	16.6	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.262	0.267	0.280	0.301	0.318	0.316	0.355	0.347	0.313	0.288	0.265	0.262	(40)	
		taud	2.632	2.637	2.570	2.483	2.461	2.476	2.457	2.470	2.548	2.607	2.639	2.641	(41)	
	Ebn at Noon		305	314	316	312	305	305	292	293	300	302	301	298	(42)	
		Edh at Noon	23	25	30	34	36	35	35	34	30	26	22	21	(43)	
(44) All-Sky Solar Radiation	RadAvg		983	1287	1764	2213	2512	2714	2394	2220	1964	1478	1104	869	(44)	
	RadStd		75	109	84	92	114	102	120	108	79	83	45	64	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (13 neighbors)		+0.88	N/A	-2.83	N/A	N/A	N/A	N/A	-211	+340	+146	

Nomenclature: See separate page